



Engr. Abdul Wahab

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RESEARCH INTERESTS

AI Mobility Systems

Autonomous and Assisted Vehicles

UAV (Unmanned Aerial Vehicle)

Connected Autonomous Vehicles (CAVs)

Vehicle Design and Engineering

Autonomous Vehicle Systems

Vehicle Dynamics and Control

PROFILE SUMMARY

My name is Abdul Wahab, and I am actively seeking an opportunity and environment where I can implement my expertise, my skills, my knowledge, and my foundation to solve the current problems. I am highly interested to learn and gain hands-on experience in above mentioned research areas because, in the coming years, tremendous advancements will happen in AI mobility. As a mechanical engineer with skills and knowledge in Autonomous systems, this direction is the right choice for me to go and become a valuable resource and contribute meaningfully to the benefits of humanity.

WORK EXPERIENCE

June 2026 — September 2026

Mechanical Design Engineer

Dysol (Pvt) Ltd, Lahore, Pakistan

R&D Department — Key Responsibilities & Contributions

- Serving as a core Mechanical Design Engineer in the Research & Development department, focusing on innovative product development.
- Leading the end-to-end design and optimization of complex mechanical and embedded systems.
- Conducting iterative engineering analysis, simulation, and validation to ensure functional performance and reliability.
- Applying DFM and DFA principles to bridge the gap between conceptual design and full-scale manufacturing.
- Collaborating across team to make sure smooth workflow and on-time delivery.

August 2025 — Present

Mechanical Design Engineer

Upwork (National & International Clients)

- Delivered professional mechanical and product design services to national and international clients, applying rigorous engineering standards.
- Engineered precise 3D CAD models, complex assemblies, 2D technical drawings, and photorealistic renders for prototyping and manufacturing.
- Optimized components for consumer products and industrial applications, focusing on optimal material selection and manufacturing processes.
- Maintained a high degree of technical communication to translate client requirements into verifiable engineering parameters.

EDUCATION & TRAINING

September 2019 — October 2023

B.Sc. Mechanical Engineering

University of Engineering and Technology (UET), Peshawar — KPK, Pakistan

FINAL YEAR THESIS PROJECT

Design and Fabrication of Mechanism for Automated Installation of 18-Liter Water Bottle for Water Dispensers

Supervised by **Prof. Usman**

Thermodynamics

Fluid Mechanics

Machine Design

CAD / CAM

Manufacturing Processes

Heat Transfer

Mechanics of Materials

Engineering Dynamics

TECHNICAL SKILLS

Mechanical Design & Engineering Simulation

- 3D CAD Modeling & Drafting (SolidWorks)
- Design for Manufacturing and Assembly (DFMA)
- Kinematics & Mechanical Systems Integration
- Finite Element Analysis (FEA) & CFD (ANSYS)
- Kinematic & Dynamic Modeling (Altair EDEM)

Autonomous Systems & Software Integration

- Advanced Driver Assistance Systems (ADAS) & Collision Avoidance
- Path-Tracking, Control Algorithms & Vehicle Dynamics Modeling
- Programming (Python, its libraries) & Simulation Environments
- Application Development (Streamlit) & Applied AI (LLMs)

COURSES & CERTIFICATIONS

- ♦ **IELTS Academic** — Band 6.0 (Listening, Reading, Writing, Speaking)
- ♦ **Certified SOLIDWORKS Associate (CSWA)** — Dassault Systèmes
- ♦ **Pakistan Engineering Council (PEC)** — Registered Engineer
- ♦ **Python & Machine Learning** — codanics.com
- ♦ **ABC of Statistics** — codanics.com
- ♦ **AI ka Chilla 2026** — codanics.com

Additional Courses

- ♦ **Lean Six Sigma Yellow Belt (LSSYB)** — udemy.com
- ♦ **Lean Six Sigma White Belt (LSSWB)** — udemy.com
- ♦ **Introduction to Minitab** — simplilearn.com
- ♦ **CPR, AED & FIRST AID** — alison.com

FEATURED PROJECTS

1. Fitness Product Development — Mechanical Dispensing System

Industry: Fitness & Consumer Products

Key Contributions:

- Designed mechanical assemblies involving gears, rack mechanisms, springs, sliding mechanisms, and functional interfaces.
- Developed solutions for multi-function operation within limited mechanical movement constraints.
- Performed multiple design iterations to improve mechanism reliability and manufacturability.
- Applied DFM and DFA principles for production-oriented design.
- Conducted motion analysis and mechanical validation.

Challenges Managed:

Optimizing complex mechanical interactions while maintaining compact design requirements. Achieved functional mechanism performance through iterative CAD refinement and simulation.

Tools Used:

SolidWorks (3D Modeling, Assembly Design, Motion Analysis), Altair EDEM (Simulation)

2. Medical Device Development — Controlled Flow Mechanical System

Industry: Medical Technology

Key Contributions:

- Designed mechanical components for controlled flow and operational safety.
- Developed assemblies involving valves, filters, actuators, and flow-control components.
- Applied mechanical design principles related to pressure management, thermal considerations, and fluid behavior.
- Implemented DFM and DFA methodologies for manufacturing readiness.
- Performed simulation-based design validation.

Challenges Managed:

Maintaining reliable mechanical operation under strict performance requirements. Improved design reliability through engineering analysis and simulation.

Tools Used:

SolidWorks (CAD Modeling & Motion Analysis), ANSYS (Simulation & Validation)

AI FEATURED PROJECTS

Doc Agent — AI Research Assistant

[View Live App ↗](#)

An intelligent document analysis system built to make Research & Development incredibly easy. Users can upload any type of file and interactively ask questions to extract complex knowledge and information hidden within the documents instantly.

[Python](#) [Streamlit](#) [Large Language Models \(LLMs\)](#) [Document Parsing](#)

2D Autonomous Vehicle Tracker

[View GitHub ↗](#)

Developed a 2D simulation environment using Python and Pygame to model and evaluate autonomous vehicle control algorithms. Implemented a Kinematic Bicycle Model with Euler forward integration to simulate realistic vehicle physics. Engineered a Pure Pursuit Path-Tracking Controller with dynamic look-ahead distances for smooth geometric trajectory following along complex S-curves, visualized through a real-time telemetry dashboard.

[Python](#) [NumPy](#) [Pygame](#) [Kinematics](#) [Control Algorithms](#)

Adaptive Cruise Control (ACC) & AEB Simulator

[View GitHub ↗](#)

Built a dynamic longitudinal control simulator for an Adaptive Cruise Control (ACC) and Autonomous Emergency Braking (AEB) system. The control algorithm calculates the desired following gap dynamically based on velocity, and utilizes simulated radar perception with Gaussian noise to mimic real-world sensors. Calculates real-time Time-To-Collision (TTC) to engage emergency braking when a collision is imminent, ensuring vehicle safety against a dynamic lead vehicle.

[Python](#) [ADAS](#) [Mathematical Modeling](#) [Pygame](#) [Collision Avoidance](#)

LANGUAGES

Language	Level
English	Proficient (B2)

Language	Level
Urdu	Fluent (C2)
Seraiki	Native (C2)

ACADEMIC
REFERENCES

Prof. Dr. Naveed Ahmad

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